INCH-POUND
MIL-DTL-7793/1C
19 October 2012
SUPERSEDING
MIL-DTL-7793/1B
18 March 2005

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL, 28 VDC, SQUARE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

Reactivated after 19 October 2012 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

REQUIREMENTS:

1. Configuration. The meter shall meet the requirements on figure 1 and table I.

2. Tolerances. Unless otherwise specified in MIL-DTL-7793 or this specification sheet, tolerances are ±0.02 inch for two-place decimals and ±0.015 inch for three-place decimals.

3. Mounting flange. Meters with fixed flanges shall be tin plated on the front surface of the mounting flange in accordance with Class C of ASTM-B545, "Tin, Electrodeposited Coatings of," or ASTM-B339, "Pig Tin," to provide an electrical ground for the case (see Note 2 of this specification sheet).

4. Life accuracy. Meter accuracy shall be ±0.1 percent from -85 to 257 °F (-65 to 125 °C) and across the maximum and minimum input operating voltage range specified in table I.

5. Power consumption. The average power consumption at 28 volts direct current (V DC) shall not be greater than 85 milliwatts.

6. Ripple voltage. The meter shall continue to operate as specified in MIL-DTL-7793 when subjected to a cyclic peak of ripple voltage (see Note 3) of less than 2.0 VDC and the frequency-voltage coordinates on figure 2.

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6.1 Ripple voltage measurement. The ripple voltage shall be measured with a peak reading voltmeter in series with a 4.0-microfarad capacitor. The higher of the two values measured when the voltmeter is successively connected to each of two polarities shall be considered the ripple voltage.

6.2 Frequency characteristics. The frequency components of the ripple voltage shall be within the limits on figure 2 when measured as conducted interference.

7. Operating temperature range. The operating temperature range shall be from -85 to 257 °F (-65 to 125 °C).

8. Time range. Total readout of the meters shall be 9999 hours.

9. Line drop. Line drop (see Note 4) shall be less than 2 VDC.

10. Transient protection. No temporary or permanent degradation shall be produced in the meter for the input voltage and time values shown on figure 3 and figure 4 (80V and 600V transient, respectively).

10.1 Transient voltage. Transient surge voltages shall be within the limits on figure 3 when converted to their equivalent step functions. Transient spike voltages shall be within the limits shown on figure 4. They shall be not greater than ±600V nor depart from the normal wave form for more than 50 microseconds.

11. Dielectric strength test voltages. The dielectric test voltages shall be 350 V root mean square (rms) at altitude specified in MIL-DTL-7793 and 600 V rms at room ambient temperature (approximately 25 ºC).

12. Weight. The weight shall be no greater than specified in Table I.

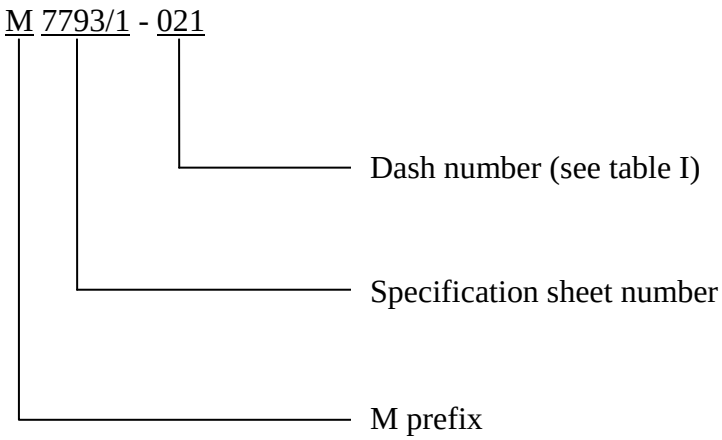
13. Qualification tests. Qualification tests shall be conducted in accordance with MIL-DTL-7793, except that the accuracy, voltage, temperature, and dielectric strength test voltages shall be as specified herein.

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NOTES:

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

1. Part or identifying number (PIN). The PIN to be used for meters acquired to this specification sheet is created as follows:



2. Meter flanges. Meters may be furnished with no flange, flange with various mounting depths, or with adjustable split mount flange in accordance with figure 1 (see MS3345-2).

3. Ripple voltage. The cyclic variation of voltage about the mean level of the DC voltage during steady-state DC electric system operation.

4. Line drop. The voltage difference between the point of voltage regulation and the power input terminals of the meter.

5. Supersession data. The meters covered by dash numbers M7793/1-029 through -054 have been superseded by the reactivated MS21341 specification sheet.

6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

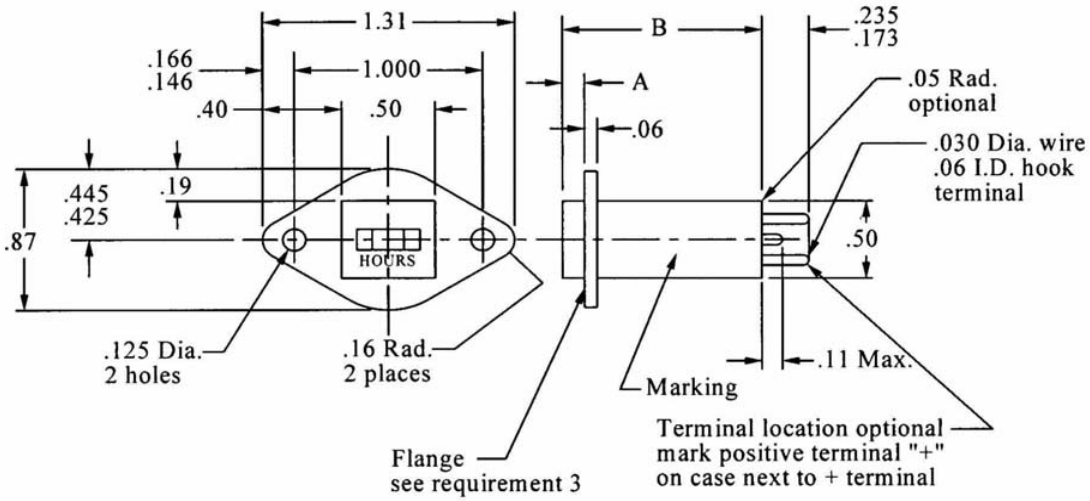
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TABLE I. Meter length, weight, and input operating voltage.

Dash No. 1/	A Flange data (inches)	B Case length (inches)		Maximum weight 2/ (ounces)	Input operating voltage	
		Minimum	Maximum		Maximum	Minimum
001	Adjustable Per MS3345-2	1.047	1.094	0.6	34	10
002	No flange					
003	0.000(flush)					
004	0.031					
005	0.062					
006	0.094					
007	0.125					
008	0.156					
009	0.188					
010	0.218					
011	0.250					
012	0.281					
013	0.312					
014	0.344					
015	0.375					
016	0.406					
017	0.438					
018	0.469					
019	0.500					
020	0.531					
021	0.562					
022	0.594					
023	0.625					
024	0.656					
025	0.688					
026	0.719					
027	0.750					
028	0.781					

1/ Maximum weight is measured with flange.

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NOTES:
1. All dimensions are in inches.
2. For dimensions A and B, see table I.

FIGURE 1. Meter configuration.

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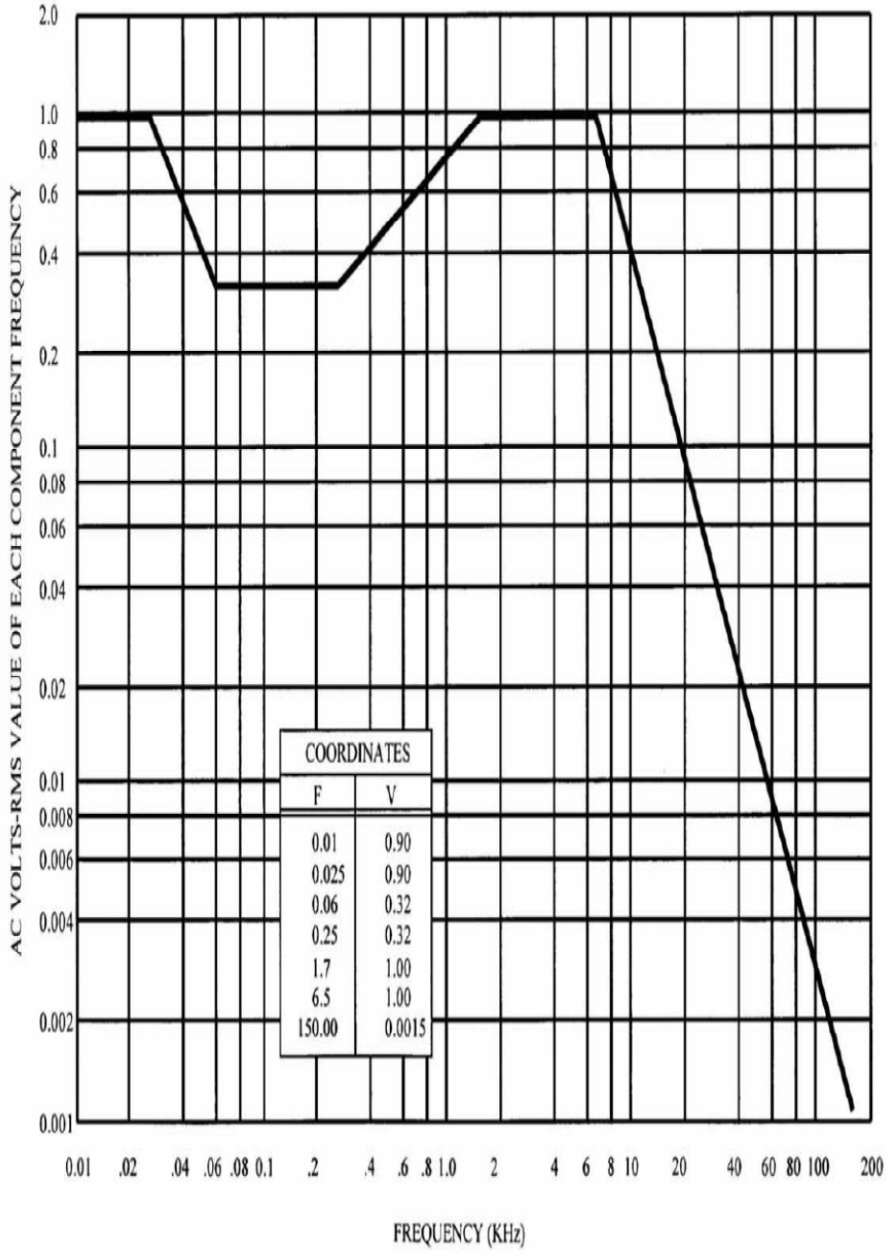


FIGURE 2. Frequency characteristics of ripple in 28 VDC.

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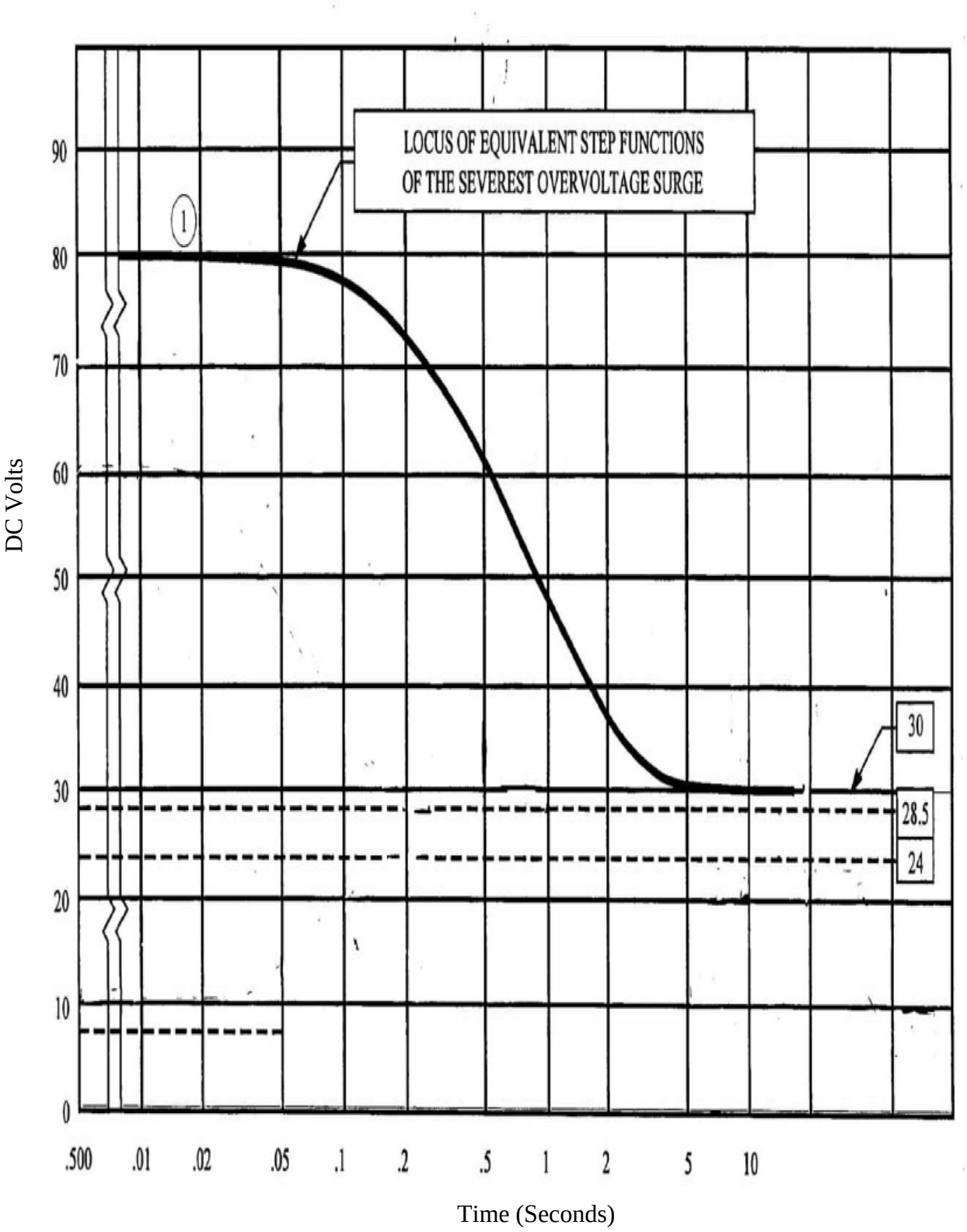


FIGURE 3. Transient surge step function loci limits.

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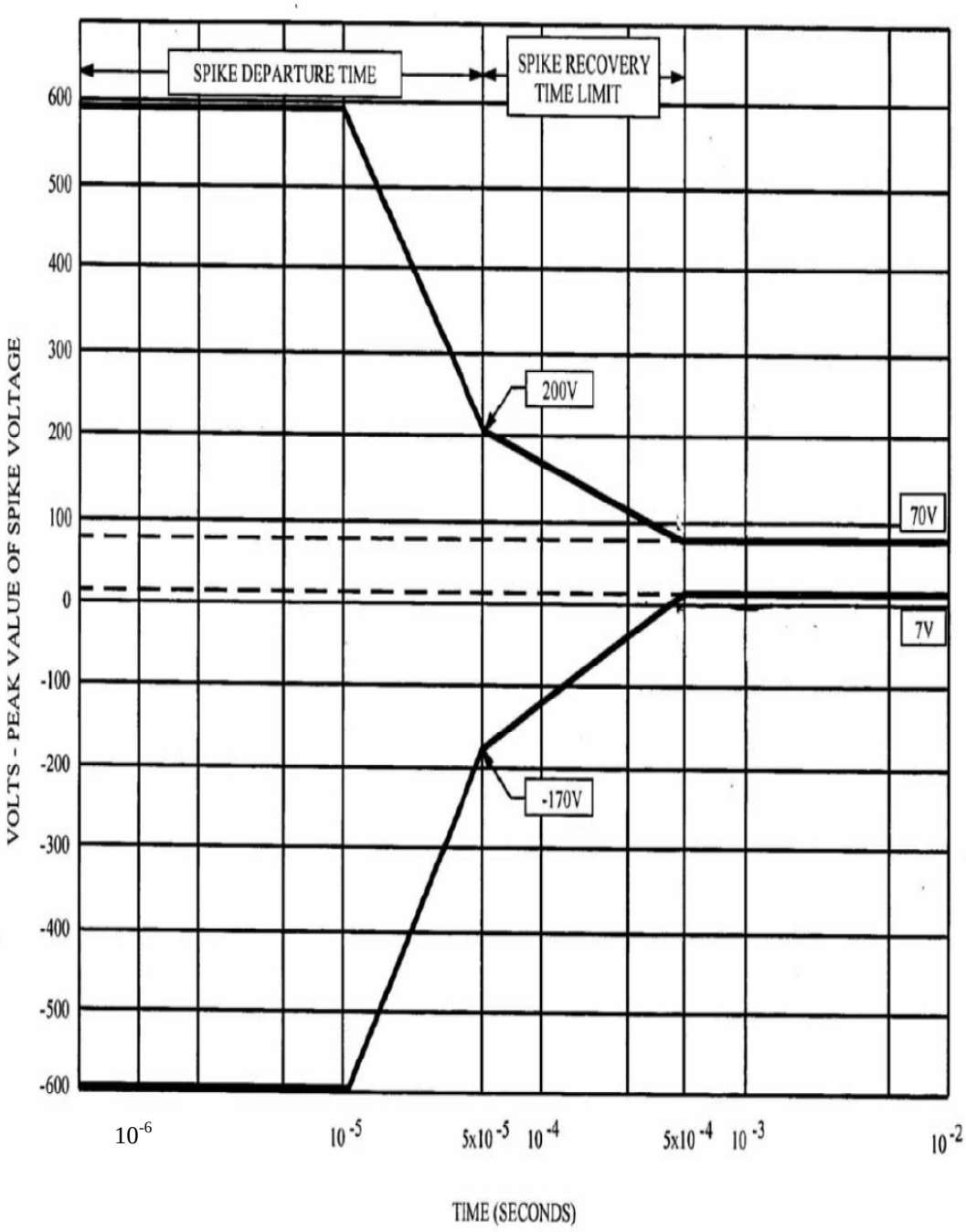


FIGURE 4. Envelope of spike voltages for direct current equipment.

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Referenced documents. In addition to MIL-DTL-7793, this specification sheet references MS3345, ASTM-B339 and ASTM-B545.

Custodians:
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS
(Project 6645-2012-003)

Review activities:
Air Force - 71
DLA - GS1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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